PECH Group Holdings Ltd

PECH: Africa’s Commerce & Infrastructure Operating System

₦250M Capital Deployment Plan — 18-Month Operational Roadmap

Prepared by: PECH Group Holdings Ltd **Date:** March 2026 **Capital Requirement:** ₦250,000,000 (Two Hundred and Fifty Million Naira) **Deployment Period:** 18 Months (March 2026 — August 2027) **Confidentiality:** This document is strictly confidential

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1. Executive Summary

PECH Group Holdings Ltd is building **Africa’s first vertically-integrated Commerce & Infrastructure Operating System** — a multi-platform ecosystem combining:

- **Embedded ERP** — Free business management software for African SMEs
- **IoT & Smart Systems Platform** — Hardware + cloud for energy, security, estate management
- **Smart Water Controllers** — Multi-tier water level controllers (Basic → IoT → Industrial/SCADA)
- **Smart Home & Electrical** — Switches, sockets, cameras (smart and non-smart), energy storage
- **Learning Tablets & EdTech** — Hardware for education content deployment with open API for creators

- **Yiwu-Style Marketplace** — Digital twin of Nigeria's physical wholesale markets
- **Spoke-and-Hub Logistics** — Cainiao-inspired parcel management platform
- **Payments Infrastructure** — PSSP/PTSP-licensed payment processing across all platforms
- **Advertising & LED Displays** — Digital signage and advertising infrastructure (Phase 2)
- **Self-Service Kiosks** — Multi-purpose payment and information kiosks

Our Mission

PECH is a **technology and platform infrastructure enabler for people**. We build the digital backbone that African businesses, homes, and communities run on. Every product we ship — from a ₦1,500 premium socket to a ₦350,000 estate system — represents **reliability, quality, and good price for your money**. We sell high-end products AND equally good affordable products, but **never bad products**. Quality is our brand promise.

Designed for Africa, Powered by PECH

All PECH products are **designed specifically for African conditions** — wide voltage tolerance (140–280V), heat resistance, dust/water protection, and offline capability. Every product is **progressively integrated across PECH's entire ecosystem**: the IoT platform powers device intelligence, the ERP connects business operations, the Marketplace enables discovery and sales, Logistics handles delivery, and Payments processes transactions. A PECH smart switch is not just a switch — it's a node in Africa's commerce and infrastructure operating system.

This plan details the deployment of **₦250,000,000** in secured capital across the first **18 months** of simultaneous lean platform launches, targeting **Nigeria first** with a clear path to West African expansion. Additional capital will be injected by the investor as the business gains traction.

Why PECH Will Win

Competitive Edge	Description
Ecosystem Lock-In	**Free ERP creates daily dependency; hardware deepens it**
African-Optimized	Offline-first, power-resilient, multi-network design
Data Moat	Transaction + energy + logistics data enables credit scoring
Founder Advantage	Founder directly sources hardware from China (Shenzhen)
No Single-Point Competition	No African company combines all 6 layers

Key Targets (18 Months)

Metric	Target
Active Merchants (ERP)	**1,500–3,000**
IoT Devices Deployed	500–1,000
Markets Digitized	3–5 major markets
Monthly Recurring Revenue	₦8M–₦15M by Month 18
Team Size	25–35 people

Payment Volume (GMV)	₦500M+ cumulative

2. The Opportunity

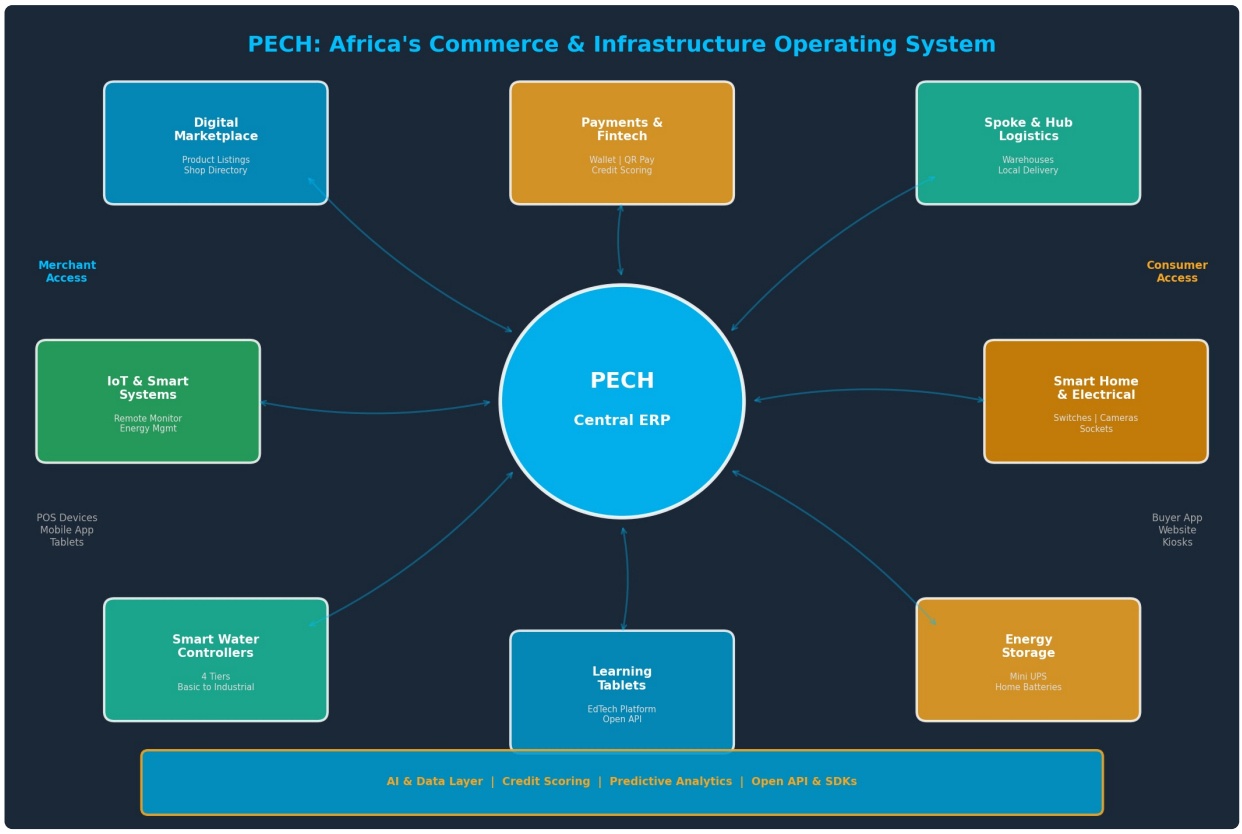
Africa’s Commerce Infrastructure Gap

Problem	Scale
African SMEs without ERP/digital tools	**90%+**
Nigerian businesses using informal credit	80%+
African IoT market (2026)	$4.85 Billion
Nigeria food waste from cold chain failure	$2.3–3.3 Billion/year
Nigerian merchants on WhatsApp-only sales	70%+
Markets with zero digital presence	Thousands

Why Now

- Smartphone penetration accelerating across Nigeria
- Digital payments adoption (Paystack, Flutterwave, OPay growing rapidly)
- CBN pushing financial inclusion and PSSP licensing
- No dominant African commerce OS exists yet — **first-mover advantage is real**
- Nigeria’s Data Protection Act (NDPA) favors local platforms

3. PECH Ecosystem Overview



PECH Commerce OS Architecture

PECH is modeled after **Alibaba Group's layered architecture**, adapted for African realities:

PECH Layer	Alibaba Equivalent	Function
Embedded ERP	**Alibaba Merchant Tools / DingTalk**	**Free business software → creates daily dependency**
YiwuGou Marketplace	1688.com / YiwuGou	Digital twin of physical markets → national discovery
Payments & Wallets	Alipay / Ant Group	Payment processing → data visibility
Logistics Platform	Cainiao Network	Spoke-and-hub parcel management → movement coordination
Fintech & Credit	Ant Financial / MYbank	Credit scoring from real data → monetization
Data & APIs	Alibaba Cloud / Data Services	Trade intelligence → sold to banks/governments
IoT & Smart Systems	Alibaba Cloud IoT	Energy, security, estate management → infrastructure
Smart Hardware	Ant POS / Alibaba Devices	POS, meters, controllers → hardware monetization

4. Ecosystem Architecture Diagram

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│                    PECH GROUP HOLDINGS LTD                               │
│              Africa's Commerce & Infrastructure OS                       │
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│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐ │
│  │   EMBEDDED   │  │  MARKETPLACE │  │  SPOKE & HUB │  │  IoT &       │ │
│  │     ERP      │  │  (YiwuGou)   │  │  LOGISTICS   │  │  SMART       │ │
│  │              │  │              │  │              │  │  SYSTEMS     │ │
│  │ • Inventory  │  │ • Market     │  │ • Spoke App  │  │              │ │
│  │ • Sales      │  │   Digitize   │  │   (52 scrns) │  │ • Energy     │ │
│  │ • Debtors    │  │ • Shop       │  │ • Hub App    │  │ • Security   │ │
│  │ • Reporting  │  │   Directory  │  │   (94 scrns) │  │ • Estates    │ │
│  │ • Mobile-1st │  │ • Discovery  │  │ • Sorting    │  │ • Meters     │ │
│  └──────┬───────┘  └──────┬───────┘  └──────┬───────┘  └──────┬───────┘ │
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│  │              PAYMENTS & FINANCIAL INFRASTRUCTURE                   │  │
│  │                                                                    │  │
│  │   ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────────┐     │  │
│  │   │ Wallet   │  │ Bank     │  │ Escrow   │  │ Smart POS    │     │  │
│  │   │ Ledger   │  │ Transfer │  │ (B2B)    │  │ Devices      │     │  │
│  │   └──────────┘  └──────────┘  └──────────┘  └──────────────┘     │  │
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│                               │                                          │
│  ┌────────────────────────────┴───────────────────────────────────────┐  │
│  │              DATA, AI & CREDIT SCORING ENGINE                      │  │
│  │                                                                    │  │
│  │  Transaction Data + Energy Usage + Payment History + Device        │  │
│  │  Uptime → Credit Score (0-100) → Banks / MFIs / Insurers          │  │
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│                               │                                          │
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│  │              OPEN API & DEVELOPER ECOSYSTEM                        │  │
│  │                                                                    │  │
│  │  REST APIs • Webhooks • SDKs (JS, Python, Flutter) • White-Label  │  │
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│  ┌────────────────────────────────────────────────────────────────────┐  │
│  │              SMART HARDWARE LAYER                                  │  │
│  │                                                                    │  │
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│  │  │Smart POS│ │Tablets │ │IoT     │ │Smart   │ │Logistics     │   │  │
│  │  │Terminal │ │& Mini  │ │Control-│ │Meters  │ │Scanners &    │   │  │
│  │  │(ERP+Pay)│ │  PC    │ │  lers  │ │(Energy)│ │POS Devices   │   │  │
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Technology Stack Architecture

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│                    DEVICE LAYER                          │
│  ESP32 • STM32 • Android POS • SUNMI Terminals           │
│  MQTT • HTTP • Modbus • LoRaWAN • NB-IoT                 │
├──────────────────────────────────────────────────────────┤
│                 EDGE INTELLIGENCE                        │
│  Offline-first • Local rules • Data buffering            │
│  NanoMQ (MIT) • Node-RED (Apache 2.0)                    │
├──────────────────────────────────────────────────────────┤
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│                    MESSAGING LAYER                          │
│  EMQX (Apache 2.0) • Apache Kafka • WebSockets              │
├─────────────────────────────────────────────────────────────┤
│                    CORE BACKEND                             │
│  Node.js / Go • PostgreSQL • TimescaleDB • Redis            │
│  ThingsBoard (Apache 2.0) • Keycloak (Auth)                 │
├─────────────────────────────────────────────────────────────┤
│                     AI ENGINE                               │
│  PyTorch • XGBoost • MLflow • FastAPI • Feast               │
├─────────────────────────────────────────────────────────────┤
│                 FRONTEND & MOBILE                           │
│  React (Web) • Flutter (Mobile) • Leaflet (Maps)            │
├─────────────────────────────────────────────────────────────┤
│               INFRASTRUCTURE & DEVOPS                       │
│  Docker • k3s (Kubernetes) • Traefik • GitHub Actions│
│  Pech Cloud (Nigeria) • Self-Host Option                    │
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5. Platform Breakdown & Build Strategy

5.1 Embedded ERP Platform

Aspect	Detail
Purpose	**Free business management for SMEs**
Target Users	Market traders, wholesalers, distributors
Core Modules	Inventory, Sales, Debtors/Creditors, Reporting
Build Strategy	Fork ERPNext (Apache 2.0) → Simplify → Mobile-first → Embed payments
Key Differentiator	Works offline, Nigerian market-optimized, integrated with payments
Revenue Model	FREE (monetize via hardware + payments + data)

5.2 YiwuGou-Style Marketplace

Aspect	Detail
Purpose	**Digital twin of Nigeria's physical markets**
Target Users	Buyers searching wholesale goods nationally
Core Features	Market → Section → Shop Directory, Verified listings, WhatsApp contact
Build Strategy	Custom build (React + Flutter + PostgreSQL)
Pilot Markets	Alaba (Lagos), Ariaria (Aba), Sabon Gari (Kano)
Revenue Model	Featured listings, premium visibility, data services

5.3 Spoke-and-Hub Logistics Platform

Aspect	Detail

Purpose	**Parcel pickup station & hub management (Cainiao model)**
Target Users	Station operators, hub managers, couriers
Core Features	Package check-in/out (52 screens), Hub management (94 screens, 355+ features), Tracking, Sorting
Build Strategy	Custom build from complete documentation spec
Hardware	SUNMI POS terminals, barcode scanners, sorting machines
Revenue Model	Per-package fees, station licensing, premium features

5.4 IoT & Smart Systems Platform

Aspect	Detail
Purpose	**Device management, energy monitoring, smart estates**
Target Users	Solar installers, estate managers, commercial buildings
Core Features	Device provisioning, OTA updates, dashboards, AI analytics
Build Strategy	Fork ThingsBoard (Apache 2.0) → Customize → Brand as Pech Cloud
Priority Verticals	Energy monitoring, Surveillance, Estate billing
Revenue Model	Hardware margin (30-40%) + SaaS (₦1,500-₦7,500/device/month)

5.5 Payments Infrastructure

Aspect	Detail
Purpose	**Process payments across ALL PECH platforms**
Strategy Phase 1	Partner with Paystack/Flutterwave (no license needed)
Strategy Phase 2	Obtain PSSP license from CBN (Month 12-18)
Strategy Phase 3	PTSP license for POS terminal operations (post-18 months)
Revenue Model	Transaction fees (0.5-1.5% per transaction)

5.6 Smart Water Controllers

Aspect	Detail
Purpose	**Multi-tier smart water level controllers for African conditions**
Target Users	Households, estates, hotels, hospitals, industrial boreholes
Tier 1 — Basic	Float switches + Arduino Nano, auto pump control, dry-run protection, LED/buzzer (₦15K–₦25K)
Tier 2 — Mid-Range	Ultrasonic sensor + ESP8266, LCD display, WiFi monitoring, dual-tank support (₦35K–₦50K)
Tier 3 — Smart IoT	ESP32 + SIM800L GSM, mobile app, SMS/USSD alerts, OTA

	firmware, energy tracking, anti-cycling (₦55K–₦80K)
Tier 4 — Industrial/SCADA	STM32 + SIM7600 4G, RS485/Modbus, LoRa, 3-phase, redundant sensors, DIN-rail (₦120K–₦200K)
ODM Strategy	Full design ownership, editable source files, NDA with Chinese ODMs, EVT→DVT→PVT validation
Key Safety	140–280VAC tolerance, MOV+TVS protection, hardwired dry-run interlock, conformal coating, IP65+
Revenue Model	Hardware margin (40-60%) + IoT SaaS for Tier 3-4

5.7 Smart Home & Electrical Products

Aspect	Detail
Purpose	**Smart and non-smart electrical products for homes and businesses**
Product Lines	Smart switches, smart sockets, smart cameras, IP cameras, non-smart switches/sockets
Phase 1 Products	Smart switches, smart sockets, WiFi cameras, basic switches/sockets
Integration	All smart products connect to PECH IoT platform, controlled via app
Quality Standard	High-end AND affordable options — **no bad products** — reliability is the brand
Revenue Model	Hardware margin (35-50%) + SaaS for camera/monitoring subscriptions

5.8 Learning Tablets & EdTech Platform

Aspect	Detail
Purpose	**Deploy educational content via PECH-branded tablets**
Phase 1 Strategy	Source Android learning tablets from China; sell as-is with pre-loaded content
Phase 2+ Strategy	Control the firmware and software; build content delivery platform
Content Model	Partner with content creators; provide **open API** for integration to PECH devices
API Strategy	Content creators get REST API + SDKs to deploy their content to PECH tablet fleet
Target Users	Schools, parents, tutoring centers, corporate training
Revenue Model	Hardware margin + content creator revenue share (30/70 split) + institution licensing

5.9 Energy Storage Systems

Aspect	Detail

Purpose	**Battery backup systems for IoT devices and home/business power**
Products	Mini UPS for IoT, LiFePO4 home batteries, solar battery systems
Integration	Monitored via PECH IoT platform, smart charging, remote diagnostics
Target Users	IoT installations, homes, small businesses, solar setups
Revenue Model	Hardware margin (30-40%) + monitoring SaaS

5.10 Advertising & LED Displays (Phase 2)

Aspect	Detail
Purpose	**Digital signage and advertising infrastructure in markets and public spaces**
Phase 2 Launch	Start with LED display hardware sourcing + advertising management platform
Products	Indoor/outdoor LED screens, digital menu boards, market advertising panels
Integration	Content managed via PECH platform, remote update, scheduling
Revenue Model	Hardware sales + advertising revenue share + display-as-a-service

6. Phased Execution Roadmap (18 Months)

18-Month Phased Roadmap

Phase 1: Foundation (Months 0–3) — “Build the Engine”

Activity	Deliverable

Core team hiring (5-7 people)	**Architecture locked, team operational**
ERP MVP development	Basic inventory + sales + mobile app
IoT platform setup	ThingsBoard fork + Pech branding
Hardware sourcing (China)	First batch POS + IoT devices ordered
Company registration & compliance	CAC, NCC engagement, legal setup
Marketplace design	UI/UX complete, backend started
Logistics app design finalization	Spoke App development started

Budget Allocation: ₦55,000,000

Phase 2: MVP & Pilot (Months 4–6) — "First Contact with Market"

Activity	Deliverable
ERP pilot (50 merchants)	**Live in 1 market (Alaba, Lagos)**
IoT pilot (30 devices)	3 solar installer partnerships
Marketplace alpha	1 market digitized (500 shops)
Spoke App alpha	2 pickup station pilots
Hardware delivery (China → Nigeria)	100 POS + 50 IoT devices
Payment integration (Paystack)	Live transactions
Field agent recruitment	10 agents deployed

Budget Allocation: ₦52,000,000

Phase 3: Market Entry (Months 7–9) — "Prove the Model"

Activity	Deliverable
ERP expansion (500 merchants)	**2 additional markets**
IoT commercial launch	Energy vertical live
Marketplace expansion	3 markets, 1,500 shops
Logistics platform beta	5 spoke stations operational
PSSP license application filed	CBN engagement started
Marketing campaigns	Lagos + Abuja
Support team scaled	3 support reps

Budget Allocation: ₦55,000,000

Phase 4: Scale & Revenue (Months 10–12) — "Revenue Engine On"

Activity	Deliverable
ERP at 1,500 merchants	**Daily active usage proven**

IoT at 300 devices	SaaS revenue flowing
Marketplace at 3,000 shops	Featured listings revenue
Logistics at 10 stations	Package volume growing
Payments processing ₦100M+/month	Transaction fee revenue
Data analytics dashboard	Credit scoring prototype
Second hardware batch	200 POS + 100 IoT devices

Budget Allocation: ₦45,000,000

Phase 5: Ecosystem Maturity (Months 13–18) — "Lock-In & Monetize"

Activity	Deliverable
ERP at 3,000 merchants	**Market dependency established**
IoT at 800 devices	Multi-vertical (energy + estates)
Marketplace at 5+ markets	National visibility
PSSP license obtained	Own payment processing
Developer API launched	SDK + webhooks live
Credit scoring pilot	2 MFI partnerships
Installer certification program	20 certified installers
Third hardware batch + new SKUs	Smart meters, tablets
West Africa research	Ghana/Kenya assessment

Budget Allocation: ₦43,000,000

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7. Capital Deployment Plan

7.1 Total Budget Allocation (₦250,000,000 / 18 Months)

Category	Amount (₦)	% of Total
Hardware Procurement (China Sourcing — 23 SKUs)	**₦75,000,000**	**30.0%**
Human Capital (Salaries & Contractors)	₦70,000,000	28.0%
Technology Infrastructure & AI Tools	₦24,000,000	9.6%
Marketing & Customer Acquisition	₦22,000,000	8.8%
Contingency & Emergency Reserve (FX + unforeseen)	₦21,000,000	8.4%
Office, Equipment & Operations	₦16,000,000	6.4%

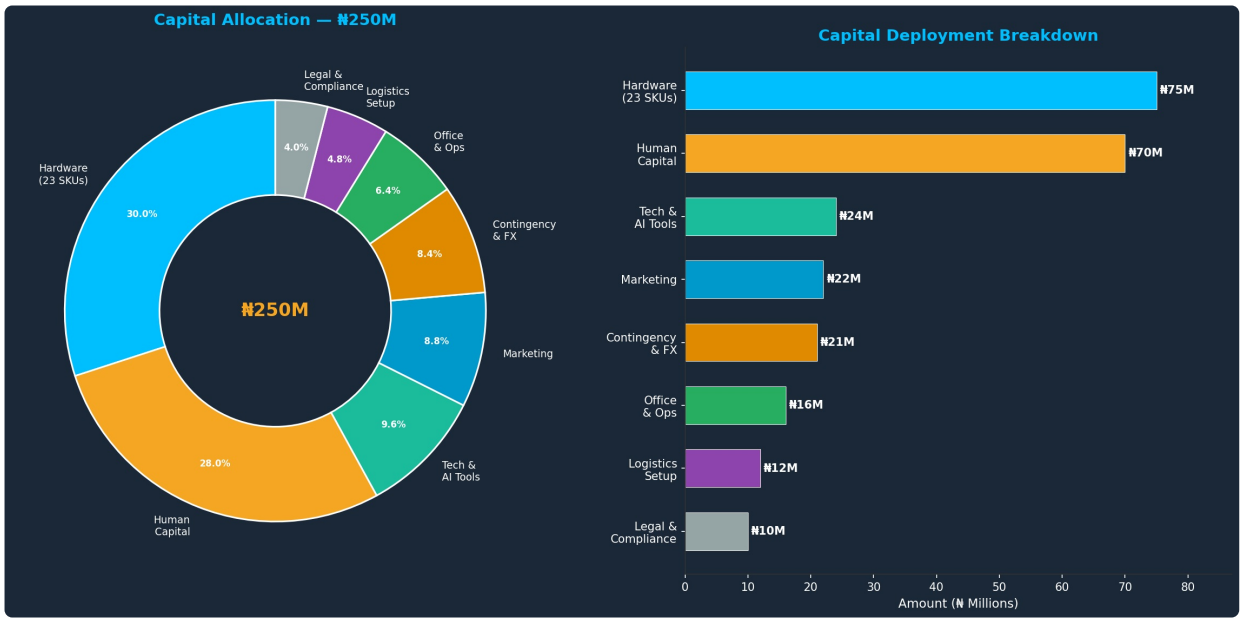
Logistics Platform (Stations Setup)	₦12,000,000	4.8%
Licensing, Legal & Compliance	₦10,000,000	4.0%
TOTAL	**₦250,000,000**	**100%**

FUNDING GAPS (Covered by investor's next tranche): - PSSP License: ~₦205M (₦100M share capital + ₦100M escrow + fees + PCI DSS) - PTSP License: ~₦201M (₦100M share capital + ₦100M escrow + fees) - Payment processing in Phase 1–3 uses licensed PSP partners (Paystack/Flutterwave) - Own PSSP/PTSP licenses pursued when investor injects additional capital

7.2 Monthly Cash Flow Projection

Quarter	Months	Monthly Burn (₦)	Cumulative Spend (₦)	Revenue (₦)
Q1	**1–3**	**₦14.0M**	**₦42.0M**	**₦0**
Q2	4–6	₦16.5M	₦91.5M	₦1.5M
Q3	7–9	₦17.0M	₦142.5M	₦8.0M
Q4	10–12	₦15.0M	₦187.5M	₦22.0M
Q5	13–15	₦12.5M	₦225.0M	₦32.0M
Q6	16–18	₦8.3M	₦250.0M	₦42.0M

Break-even projected: Month 14–16 (operational break-even)



Budget Allocation

7.3 Budget Allocation Visual

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Hardware (23 SKUs)  ▓▓▓
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Contingency/FX      ▓▓▓
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	Manager			

Monthly payroll: ₦1,950,000 **Quarterly payroll:** ₦5,850,000

> Compliance handled by legal retainer (₦100K/mo). UI/UX contracted per-project basis.

Phase 2 (Months 4–6): MVP & Pilot — 9 People

#	Role	Monthly Salary (₦)	Notes
6	**IoT/Embedded Engineer**	**₦400,000**	**Firmware + device integration + water controllers**
7	**DevOps Engineer**	₦350,000	Cloud infrastructure + CI/CD
8–9	**Field Agents (2 contract)**	₦80,000 each + per-merchant bonus	Merchant acquisition

Monthly payroll: ₦3,060,000 **Quarterly payroll:** ₦9,180,000

Phase 3 (Months 7–9): Market Entry — 13 People

#	Role	Monthly Salary (₦)	Notes
10	**Payments Product Manager**	**₦600,000**	**Payment complexity management**
11	**Data Analyst**	₦350,000	Credit scoring + insights
12–13	**Support Representatives (2)**	₦150,000 each	Scale merchant support

Monthly payroll: ₦4,310,000 **Quarterly payroll:** ₦12,930,000

Phase 4 (Months 10–12): Revenue Scale — 18 People

#	Role	Monthly Salary (₦)	Notes
14	**Logistics Operations Lead**	**₦350,000**	**Spoke-and-hub management**
15–16	**Junior Engineers (2)**	₦250,000 each	Scale development
17–18	**Additional Field Agents (2)**	₦80,000 each	2nd/3rd market

Monthly payroll: ₦5,320,000 **Quarterly payroll:** ₦15,960,000

Phase 5 (Months 13–18): Ecosystem Maturity — 22-25 People

#	Role	Monthly Salary (₦)	Notes
	Senior Data		

19	**Engineer**	**₦500,000**	**AI/ML pipeline**
20	**Business Development Lead**	₦400,000	Enterprise + partnerships
21	**Regional Field Coordinator**	₦150,000	Abuja expansion
22–25	**Additional Field Agents (4)**	₦80,000 each	National expansion

Monthly payroll: ₦6,690,000 **Quarterly payroll (6 months):** ₦40,140,000

8.2 Total Human Capital Cost (18 Months)

Phase	Duration	Avg Monthly (₦)	Total (₦)
Phase 1	**3 months**	**₦1,950,000**	**₦5,850,000**
Phase 2	3 months	₦3,060,000	₦9,180,000
Phase 3	3 months	₦4,310,000	₦12,930,000
Phase 4	3 months	₦5,320,000	₦15,960,000
Phase 5	6 months	₦6,690,000	₦40,140,000
TOTAL	**18 months**		**₦84,060,000**

Note: Actual budget is ₦70,000,000 for human capital. The difference is covered by: - Delayed hiring (Phase 5 roles shift to Month 15-18 based on revenue) - Contract/per-project arrangements (UI/UX, compliance, field agents) - Revenue offsetting costs from Month 10+ - Performance-based bonuses replacing fixed salary increases - Lean team philosophy — hire only when revenue justifies it

Salary Validation (March 2026 Market Data): All salary figures benchmarked against Glassdoor, PayScale, and Levels.fyi Nigeria data: - CTO at ₦800K/mo is **competitive** (market range: ₦400K–₦1.4M) - Engineers at ₦400K–₦450K/mo are **lean/startup-tier** (market mid: ₦500K–₦800K), offset by equity - DevOps at ₦350K/mo is **below market mid** (₦250K–₦600K) — adjust after 6 months - Junior engineers at ₦250K/mo are **reasonable** for entry-level (market: ₦150K–₦300K) - Field agents at ₦80K + bonus is **standard** for contract agents - All salaries for **on-site/local roles**; remote international rates 3–5x higher

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9. Technology & AI Tools Budget

9.1 Cloud Infrastructure

Service	Monthly Cost (₦)	Annual (₦)	Notes
Cloud Hosting (AWS/Azure Nigeria)	**₦600,000**	**₦7,200,000**	**Multi-service hosting**
Database (PostgreSQL + TimescaleDB)	₦150,000	₦1,800,000	Managed instances
MQTT Broker (EMQX Cloud)	₦100,000	₦1,200,000	IoT messaging
CDN & Storage (S3-compatible)	₦80,000	₦960,000	Static assets + backups

Domain & SSL certificates	₦20,000	₦240,000	Multiple domains

18-Month Infrastructure Total: ₦17,100,000

9.2 AI & Development Tools

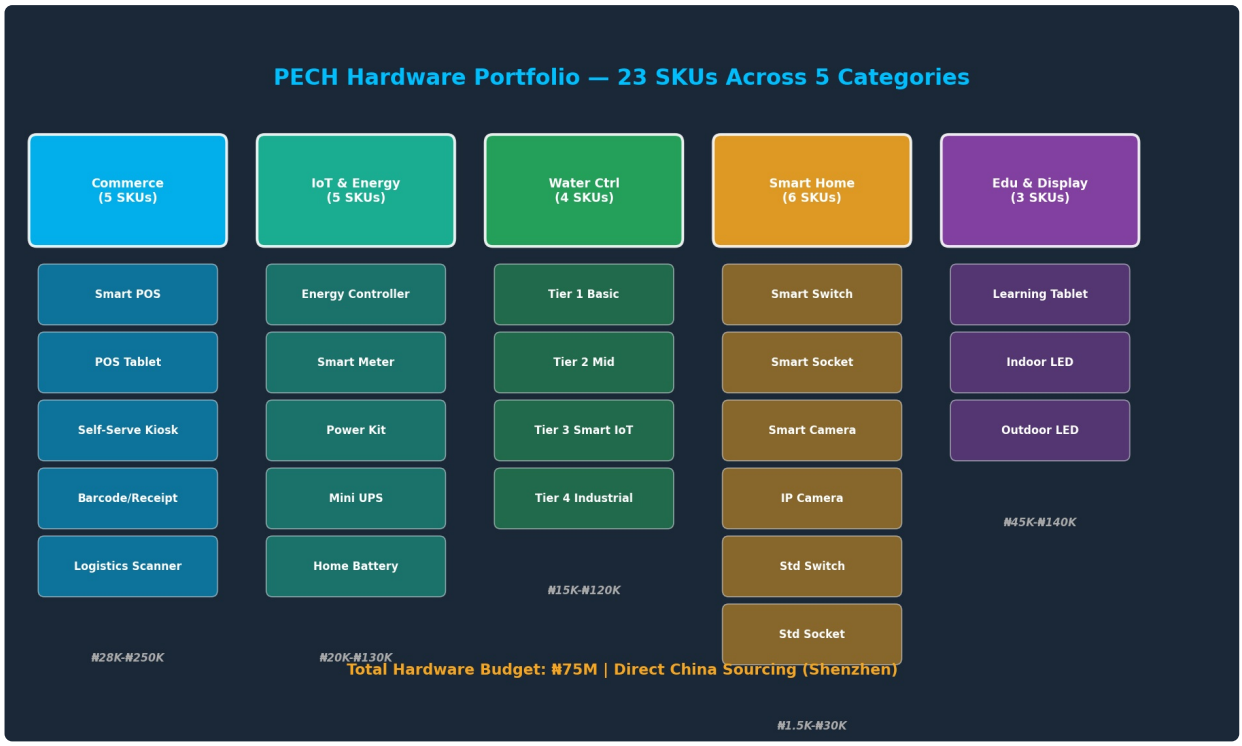
Tool	Monthly Cost (₦)	Purpose
GitHub Organization (Team)	**₦25,000**	**Code hosting + CI/CD**
Claude Pro / ChatGPT Team (5 seats)	₦150,000	AI-assisted development
Cursor IDE (5 seats)	₦100,000	AI pair programming
Figma (3 seats)	₦45,000	UI/UX design
Slack / Communication	₦30,000	Team coordination
Postman / API Tools	₦20,000	API testing
Sentry (Error monitoring)	₦30,000	Production monitoring
MLflow (Self-hosted)	₦0	ML experiment tracking

Monthly AI/Tools: ₦400,000 **18-Month Tools Total:** ₦7,200,000

9.3 Total Technology Budget

Category	18-Month Total (₦)
Cloud Infrastructure	**₦17,100,000**
AI & Development Tools	₦7,200,000
TOTAL	**₦24,300,000**

10. Hardware & Device Strategy



Hardware Product Portfolio — 23 SKUs

10.1 Hardware Product Line

Commerce & Business Devices

SKU	Device	Unit Cost (₦)	Sell Price (₦)	Margin
SKU-01	**Smart POS Terminal (Android, 4G, printer built-in)**	**₦45,000**	**₦65,000**	**44%**
SKU-02	**POS Tablet PC** (10”, Android, LTE, for merchants)	₦35,000	₦55,000	57%
SKU-03	**Self-Service Kiosk** (Touch screen, payment, info)	₦150,000	₦250,000	67%
SKU-04	**Receipt + Barcode Kit**	₦18,000	₦28,000	56%
SKU-05	**Logistics Scanner** (SUNMI-compatible)	₦25,000	₦40,000	60%

IoT & Energy Devices

SKU	Device	Unit Cost (₦)	Sell Price (₦)	Margin

SKU-06	**IoT Energy Controller (ESP32, MQTT, solar)**	**₦15,000**	**₦25,000**	**67%**
SKU-07	**Smart Meter** (energy monitoring, WiFi)	₦20,000	₦35,000	75%
SKU-08	**Power Stability Kit** (Mini UPS + surge)	₦12,000	₦20,000	67%
SKU-09	**Energy Storage — Mini UPS** (IoT backup, LiFePO4)	₦25,000	₦40,000	60%
SKU-10	**Energy Storage — Home Battery** (1kWh LiFePO4)	₦80,000	₦130,000	63%

Smart Water Controllers

SKU	Device	Unit Cost (₦)	Sell Price (₦)	Margin
SKU-11	**Water Controller Tier 1 (Basic — float switches, Arduino)**	**₦8,000**	**₦15,000**	**88%**
SKU-12	**Water Controller Tier 2** (Mid — ultrasonic, ESP8266, LCD)	₦18,000	₦35,000	94%
SKU-13	**Water Controller Tier 3** (Smart IoT — ESP32, GSM, app)	₦30,000	₦55,000	83%
SKU-14	**Water Controller Tier 4** (Industrial — STM32, 4G, SCADA)	₦65,000	₦120,000	85%

Smart Home & Electrical

SKU	Device	Unit Cost (₦)	Sell Price (₦)	Margin
SKU-15	**Smart Switch (WiFi, app control, 1-4**	**₦3,000**	**₦7,000**	**133%**

	gang)			
SKU-16	**Smart Socket** (WiFi, energy monitoring)	₦2,500	₦5,500	120%
SKU-17	**Smart Camera** (WiFi, 1080p, night vision, 2-way audio)	₦8,000	₦18,000	125%
SKU-18	**IP Camera** (PoE, outdoor, 2K)	₦15,000	₦30,000	100%
SKU-19	**Non-Smart Switch** (Premium quality, 1-4 gang)	₦800	₦2,000	150%
SKU-20	**Non-Smart Socket** (Premium quality, 13A)	₦600	₦1,500	150%

Education & Display

SKU	Device	Unit Cost (₦)	Sell Price (₦)	Margin
SKU-21	**Learning Tablet (8", Android, pre-loaded content)**	**₦25,000**	**₦45,000**	**80%**
SKU-22	**LED Display Panel** (P3 indoor, 64×32cm)	₦40,000	₦70,000	75%
SKU-23	**Outdoor LED Screen** (P5, weatherproof, 96×48cm)	₦80,000	₦140,000	75%

10.2 Hardware Procurement Schedule

Batch	Month	Devices	Total Cost (₦)	Notes
Batch 1	**Month 3**	**100 POS + 50 IoT + 30 Scanners**	**₦8,550,000**	**Pilot devices**
Batch 2	Month 7	200 POS + 100 IoT + 50 Kits	₦16,600,000	Market entry
Batch 3	Month 12	300 POS + 200 IoT + 100 Meters	₦22,500,000	Scale batch
Shipping & Import (3 batches)	Various	—	₦7,500,000	Customs + SONCAP (~₦1.5M/shipment

				for PC+SC+lab)
TOTAL		**1,130 devices**	**₦55,150,000**	

Hardware Cost Validation (2026 China Wholesale Prices): - ESP32 modules: $1.90–$4.00/unit (~₦2,850–₦6,000) — aligns with ₦15K IoT controller cost (includes PCB, enclosure, assembly) - Smart meters (WiFi): $8–$20/unit (~₦12K–₦30K) — aligns with ₦20K unit cost - LoRa gateways: $60–$200/unit (~₦90K–₦300K) — not in current SKU list but relevant for estate deployments - Android POS terminals: $30–$60/unit (~₦45K–₦90K) — aligns with ₦45K unit cost - Shipping per 20ft container: $2,000–$4,000 (~₦3M–₦6M) - SONCAP certification per shipment: $500–$2,000 (~₦750K–₦3M) for PC + SC + lab testing

10.3 Solution Packages (What We Actually Sell)

Commerce Packages

Package	Contents	Price (₦)	Target
Micro Trader Kit	**Smart POS + Receipt Printer + Free ERP**	**₦75,000**	**Kiosks, small shops**
SME Business Kit	Smart POS + Tablet + Barcode Scanner + Power Kit + ERP	₦155,000	Wholesalers
Market Power Seller Kit	2× POS + Tablet + Full Kit + Priority Support	₦280,000	Large market shops

IoT & Energy Packages

Package	Contents	Price (₦)	Target
IoT Starter Kit	**3× Energy Controllers + Smart Meter + Dashboard**	**₦120,000**	**Solar installers**
Estate Management Kit	5× Smart Meters + CCTV + Access Control + Platform	₦350,000	Estate managers
Water Management Kit	Tier 3 Water Controller + Smart Meter + App Setup	₦85,000	Households, boreholes
Industrial Water Kit	Tier 4 Controller + Pressure Sensor + 4G + SCADA Dashboard	₦180,000	Hotels, hospitals, estates

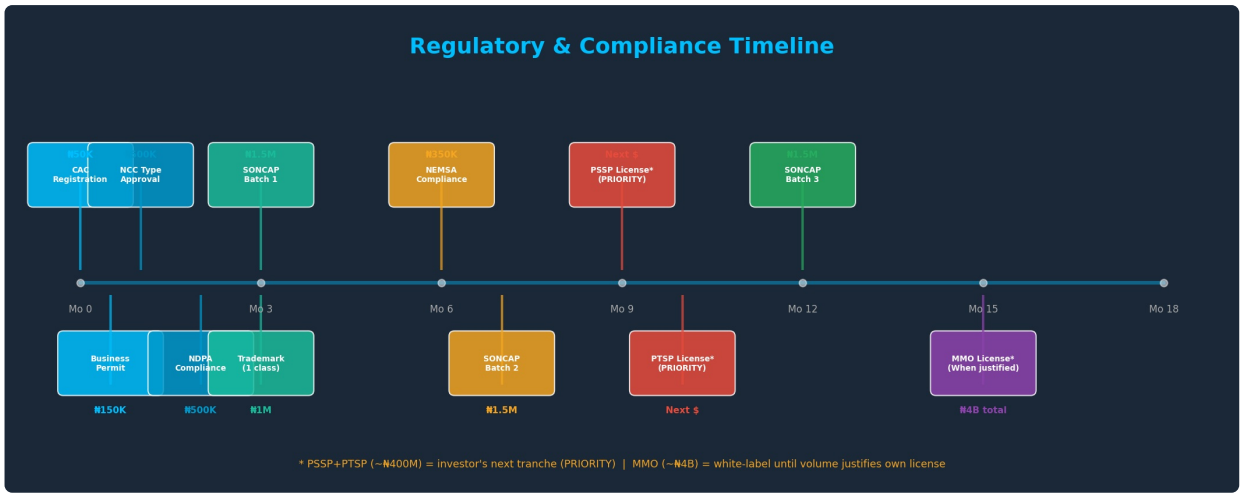
Smart Home Packages

Package	Contents	Price (₦)	Target
Smart Home Starter	**4× Smart Switches + 2× Smart Sockets + App Setup**	**₦45,000**	**Homeowners**
Home Security Kit	2× Smart Cameras + Smart Switch + Cloud Storage	₦55,000	Families, landlords
Premium Home Kit	8× Smart Switches + 4× Sockets + 2× Cameras + Energy Monitor	₦120,000	Premium homes

Education & Display Packages

Package	Contents	Price (₦)	Target
Learning Kit	**Learning Tablet + Protective Case + 1 Year Content**	**₦55,000**	**Parents, schools**
Classroom Kit	10× Learning Tablets + Teacher Dashboard + Content License	₦480,000	Schools
Digital Signage Kit	LED Display + Media Player + PECH Ad Platform	₦90,000	Shops, markets

11. Licensing, Compliance & Regulatory



Regulatory & Compliance Timeline

11.1 Required Licenses & Approvals

#	License/Approval	Authority	Cost (₦)	Timeline	Phase	From ₦250M?
1	**Company Registration (CAC)**	**CAC**	**₦50,000**	**2-4 weeks**	**Month 1**	**Yes**
2	Business Premises Permit	State Govt	₦150,000	2–3 weeks	Month 1	Yes
3	NCC Type Approval (IoT + Water Controllers)	NCC	₦500,000	3–6 months	Month 1	Yes
4	NDPA Compliance Registration + DPO	NDPC	₦500,000	4–8 weeks	Month 2	Yes
5	SONCAP (Hardware Import — PC + SC)	SON	₦1,500,000/shipment	Per batch	Month 3	Yes
6	Trademark Registration (1 class now, 2 later)	Trademarks Registry	₦1,000,000	6–12 months	Month 1	Yes
7	NEMSA Compliance (Smart Meters + Water Controllers)	NEMSA	₦350,000	2–4 months	Month 6	Yes

8	Legal Retainer (Regulatory Counsel)	Law Firm	₦100,000/month	Ongoing	Month 1	Yes
9	**PSSP License Application**	CBN	₦1,100,000	6–12 months	Month 9	**Next tranche**
10	**PSSP Share Capital**	CBN	₦100,000,000	With application	Month 9	**Next tranche**
11	**PSSP Escrow Deposit**	CBN	₦100,000,000	With application	Month 9	**Next tranche**
12	**PTSP License Application**	CBN	₦100,000	6–12 months	Month 12+	**Next tranche**
13	**PTSP Share Capital**	CBN	₦100,000,000	With application	Month 12+	**Next tranche**
14	**PTSP Escrow Deposit**	CBN	₦100,000,000	With application	Month 12+	**Next tranche**
15	PCI DSS Certification	PCI SSC	₦5,000,000	2–4 months	Month 10+	**Next tranche**
16	NERC Mini-Grid Permit (if applicable)	NERC	₦200,000	3–6 months	Month 12	**Next tranche**
17	**MMO License Application** (for wallets & escrow)	CBN	₦1,100,000	10–13 months	When justified	**Future tranche**
18	**MMO Share Capital**	CBN	₦2,000,000,000	With application	When justified	**Future tranche**
19	**MMO Escrow Deposit**	CBN	₦2,000,000,000	With application	When justified	**Future tranche**

11.2 PSSP, PTSP & MMO License — Detailed Breakdown

Item	PSSP (Payments)	PTSP (POS Terminals)	MMO (Wallets & Escrow)
Application Fee	**₦100,000**	**₦100,000**	**₦100,000**
Licensing Fee	₦1,000,000	₦1,000,000	₦1,000,000
Minimum Share Capital	₦100,000,000	₦100,000,000	₦2,000,000,000
CBN Escrow Deposit	₦100,000,000 (refundable)	₦100,000,000 (refundable)	₦2,000,000,000 (refundable)
PCI DSS Certification	₦3,000,000–₦8,000,000	Included with PSSP	Included with PSSP
Approval Process	AIP (2–3 mo) → Final (4–6 mo)	Same	AIP (6 mo) → Final (4–7 mo)
Total Capital Required	**~₦205,000,000**	**~₦201,000,000**	**~₦4,001,000,000**
What It	Payment processing,	POS terminal deployment	E-money issuance, wallets, customer

Permits	gateway	POS terminal deployment	escrow

ESCROW DURATION: All CBN escrow deposits (PSSP, PTSP, MMO) are held **for the entire duration of the license** — as long as you operate under that license. CBN invests the escrow in treasury bills. Escrow is **refundable only when you surrender the license**. Annual license reviews assess performance targets.

COMBINED PSSP + PTSP CAPITAL: If pursuing both licenses, total capital requirement is approximately **₦400M+** (₦200M PSSP + ₦200M PTSP). A single entity may hold both with shared capital — requires CBN clarification.

MMO LICENSE (FOR WALLETS & ESCROW): A PSSP license alone **does NOT permit** holding customer funds or issuing digital wallets. To hold merchant escrow, operate business wallets, or issue e-money, an **MMO license is required** (₦2B capital + ₦2B escrow = ₦4B total). Until the ₦4B is justifiable, PECH will partner with an existing licensed MMO in a **white-label arrangement** to provide wallet services across all platforms.

11.3 Payment & Wallet License Strategy

Path	Approach	Timeline	Capital Needed	Priority
Path A (Now)	**Partner with Paystack/Flutterwave for payment processing + white-label MMO for wallet services**	**Immediate (Month 1)**	**₦0**	**Phase 1**
Path B (PRIORITY)	Apply for **PSSP license** — own payment processing. Investor injects capital ASAP.	As soon as next tranche arrives	~₦205M	**HIGH — GET ASAP**
Path C (PRIORITY)	Apply for **PTSP license** — own POS terminal network. Apply alongside or shortly after PSSP.	Alongside PSSP	~₦201M	**HIGH — GET ASAP**
Path D (When justified)	Apply for **MMO license** — own wallet/escrow capability. When transaction volume justifies ₦4B capital.	When revenue justifies	~₦4B	Future

PSSP and PTSP are the #1 priority for the investor's next capital tranche. These licenses are critical for PECH to control its own payment infrastructure. The investor has committed to injecting additional capital as the business gains traction — PSSP/PTSP applications should be filed **immediately** when that capital arrives.

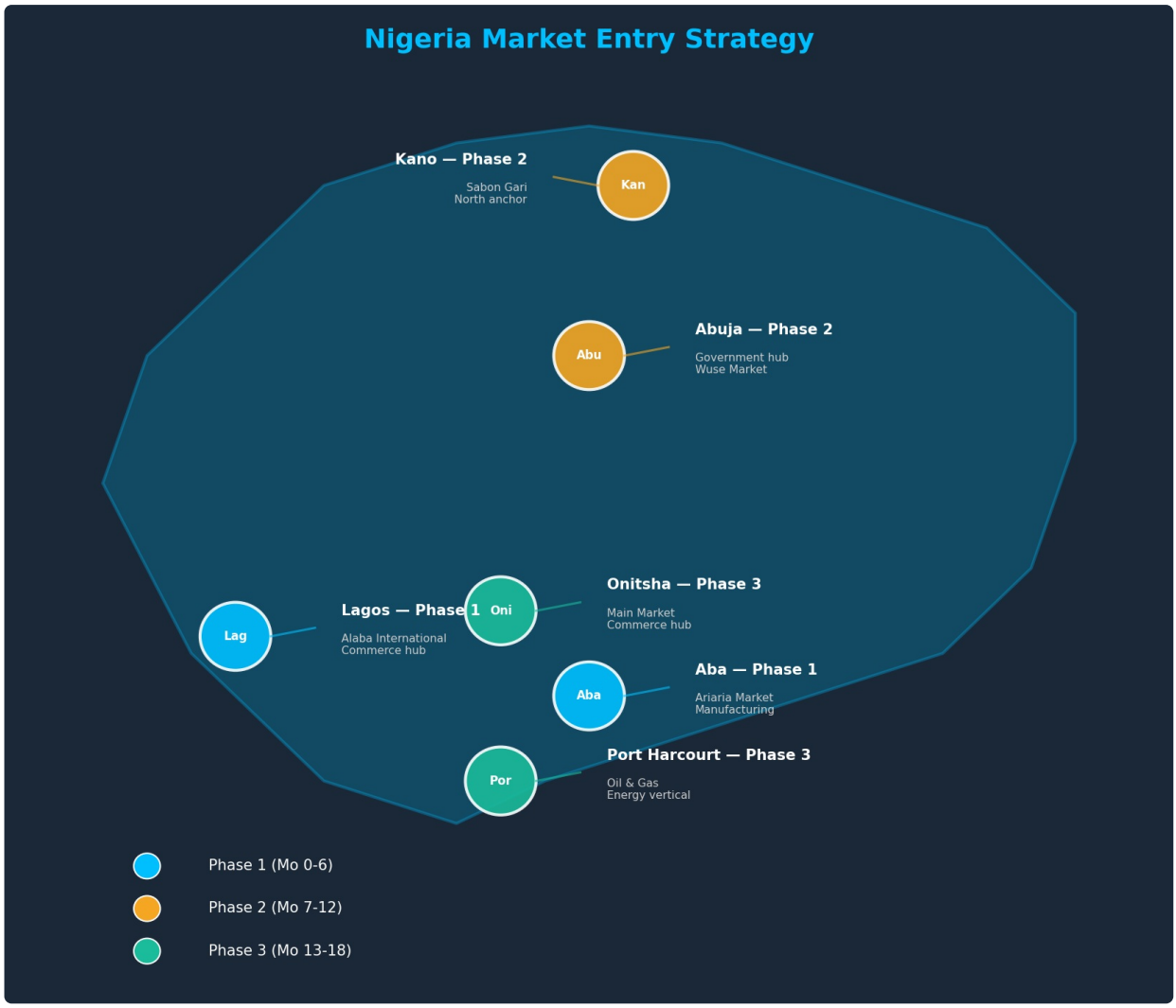
For wallets and escrow, PECH will operate under a **white-label MMO partnership** until transaction volumes and revenue justify the ₦4B capital requirement for an own MMO license.

11.4 Total Compliance Budget (From ₦250M)

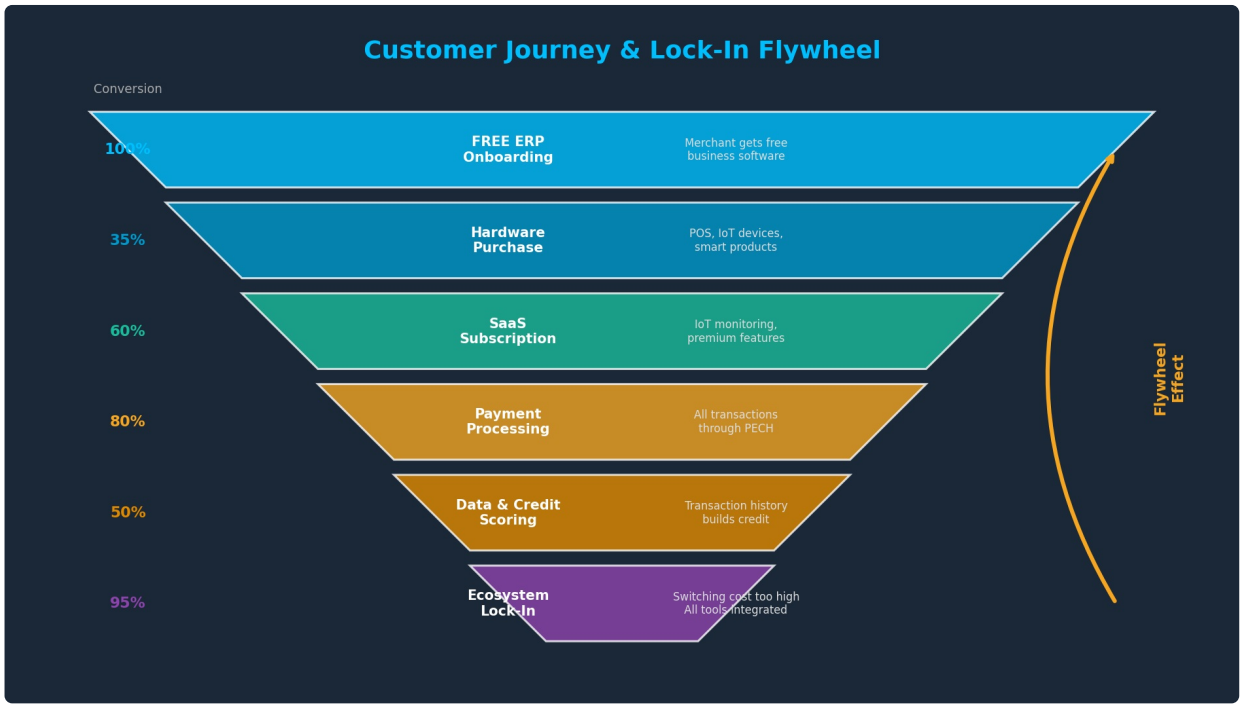
Category	Amount (₦)	Notes
Registration & Permits (CAC + Premises)	**₦200,000**	**One-time**
Device Certifications (NCC + NEMSA)	₦850,000	IoT devices + water controllers + meters

Data Protection (NDPA + DPO setup)	₦500,000	Annual CARs ongoing
Import Compliance (SONCAP × 3 shipments)	₦4,500,000	~₦1.5M per shipment (PC + SC + lab testing)
Trademarks (1 class — expand later)	₦1,000,000	Remaining 2 classes deferred to next tranche
Legal Retainer (18 months × ₦100K)	₦1,800,000	Lean retainer — scale with complexity
SUBTOTAL (From ₦250M Budget)	**₦8,850,000**	
Buffer for unexpected compliance	₦1,150,000	NCC delays, additional certifications
TOTAL COMPLIANCE (From ₦250M)	**₦10,000,000**	
DEFERRED TO INVESTOR'S NEXT TRANCHE (PRIORITY):		
PSSP License + Capital + Escrow	₦205,000,000	**PRIORITY — apply immediately when capital arrives**
PTSP License + Capital + Escrow	₦201,000,000	**PRIORITY — apply alongside or shortly after PSSP**
PCI DSS Certification	₦5,000,000	Required before own payment processing
Remaining 2 trademark classes	₦2,000,000	
NERC permit	₦200,000	If applicable
DEFERRED TO FUTURE TRANCHE (when justified):		
MMO License + Capital + Escrow	₦4,001,000,000	When transaction volume justifies; white-label MMO until then

12. Marketing & Customer Acquisition



Nigeria Market Entry Strategy



Customer Journey & Lock-In Flywheel

12.1 Marketing Strategy

Phase	Strategy	Budget (₦)
Months 1-3	**Brand identity, website, social media setup**	**₦2,000,000**
Months 4-6	Market association partnerships, demo units, pilot events	₦4,000,000
Months 7-9	Digital marketing (Google, Meta, TikTok), installer workshops	₦5,000,000
Months 10-12	Trade shows, enterprise sales collateral, case studies	₦4,000,000
Months 13-18	National campaigns, developer hackathon, brand ambassadors	₦7,000,000
TOTAL		**₦22,000,000**

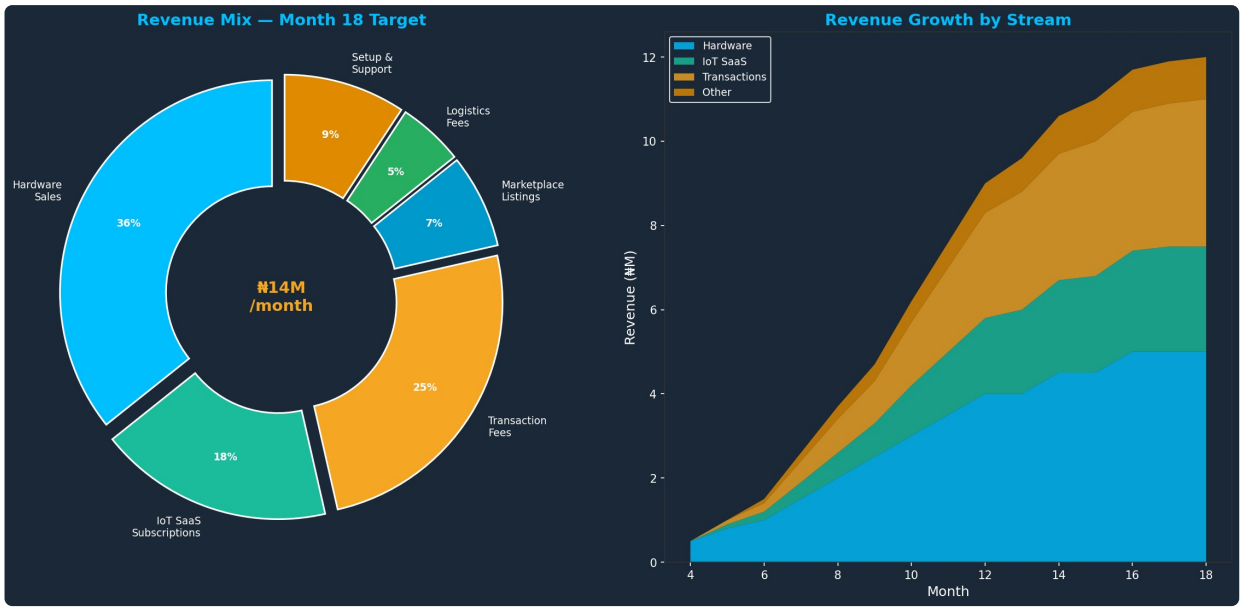
12.2 Customer Acquisition Cost Targets

Channel	CAC Target (₦)	Lifetime Value (₦)	LTV:CAC Ratio
Field Agents (ERP merchants)	**₦5,000**	**₦120,000+**	**24:1**
Digital Marketing (Marketplace)	₦2,000	₦50,000+	25:1
Installer Partnerships (IoT)	₦15,000	₦300,000+	20:1
Logistics Station Onboarding	₦25,000	₦500,000+	20:1

12.3 Market Entry Tactics (Nigeria-Specific)

Tactic	Description	Cost
Market Association MoUs	**Partnership with Alaba, Ariaria, Sabon Gari leadership**	**₦0 (relationship)**
Demo Station Setup	Physical demo in 3 target markets	₦1,500,000
Installer Certification Program	Train 20 certified PECH installers	₦1,000,000
WhatsApp Business Integration	Leverage existing merchant behavior	₦200,000
Referral Bonus Program	₦5,000 per successful merchant referral	Variable

13. Revenue Model & Financial Projections



Revenue Streams Breakdown

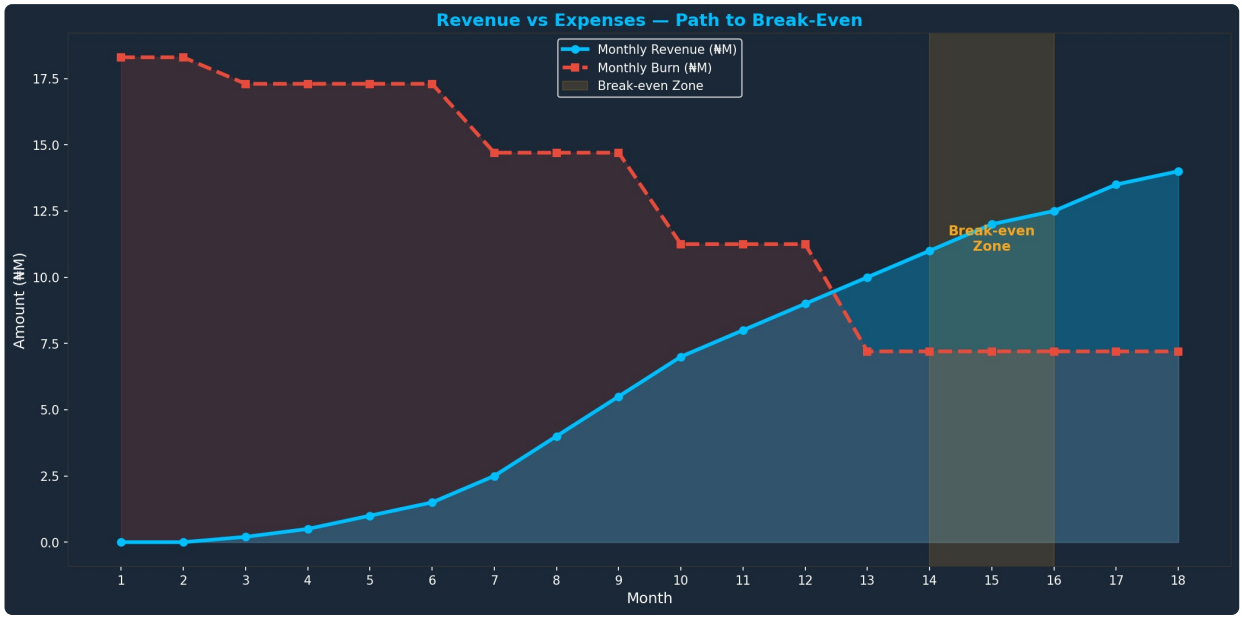
13.1 Revenue Streams

Stream	Source	Model	Monthly Target (Month 18)
Hardware Sales	**POS, IoT devices, kits**	**One-time**	**₦5,000,000**
Setup & Training	Installation fees	One-time	₦800,000
IoT SaaS	Device monitoring subscriptions	Recurring	₦2,500,000
Transaction Fees	Payment processing (0.5–1.5%)	Per-transaction	₦3,500,000
Marketplace	Featured listings, premium visibility	Recurring	₦1,000,000
Logistics	Per-package fees, station licensing	Per-transaction	₦700,000
Support Subscriptions	Premium support plans	Recurring	₦500,000
TOTAL MRR TARGET			**₦14,000,000**

13.2 18-Month Revenue Projection

Month	Hardware (₦M)	SaaS (₦M)	Transactions (₦M)	Other (₦M)	Total (₦M)
1–3	**0**	**0**	**0**	**0**	**0**
4	0.5	0	0	0	**0.5**
5	0.8	0.1	0.1	0	**1.0**

6	1.0	0.2	0.2	0.1	**1.5**
7	1.5	0.4	0.5	0.2	**2.6**
8	2.0	0.6	0.8	0.3	**3.7**
9	2.5	0.8	1.0	0.4	**4.7**
10	3.0	1.2	1.5	0.5	**6.2**
11	3.5	1.5	2.0	0.6	**7.6**
12	4.0	1.8	2.5	0.7	**9.0**
13	4.0	2.0	2.8	0.8	**9.6**
14	4.5	2.2	3.0	0.9	**10.6**
15	4.5	2.3	3.2	1.0	**11.0**
16	5.0	2.4	3.3	1.0	**11.7**
17	5.0	2.5	3.4	1.0	**11.9**
18	5.0	2.5	3.5	1.0	**12.0**
TOTAL	**46.3**	**18.5**	**27.8**	**8.5**	**₦103.6M**



Revenue vs Expenses

13.3 Path to Profitability

```
Revenue vs Expenses (₦ Millions per Month)

Month:   1   3   6   9   12   15   18

Revenue: 0   0   1.5  4.7  9.0  11.0  12.0
         ▒   ▒   ▌▒   ▌   ▌▌   ▌▌▌   ▌▌▌

Expense: 14  14  16  17   15   12.5   8.3
         ▌   ▌   ▌▌  ▌▌   ▌▌   ▌▒▒   ▌▒▒▒
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✦ Break-even: Month 14–15
✦ Cash-positive: Month 16+
```
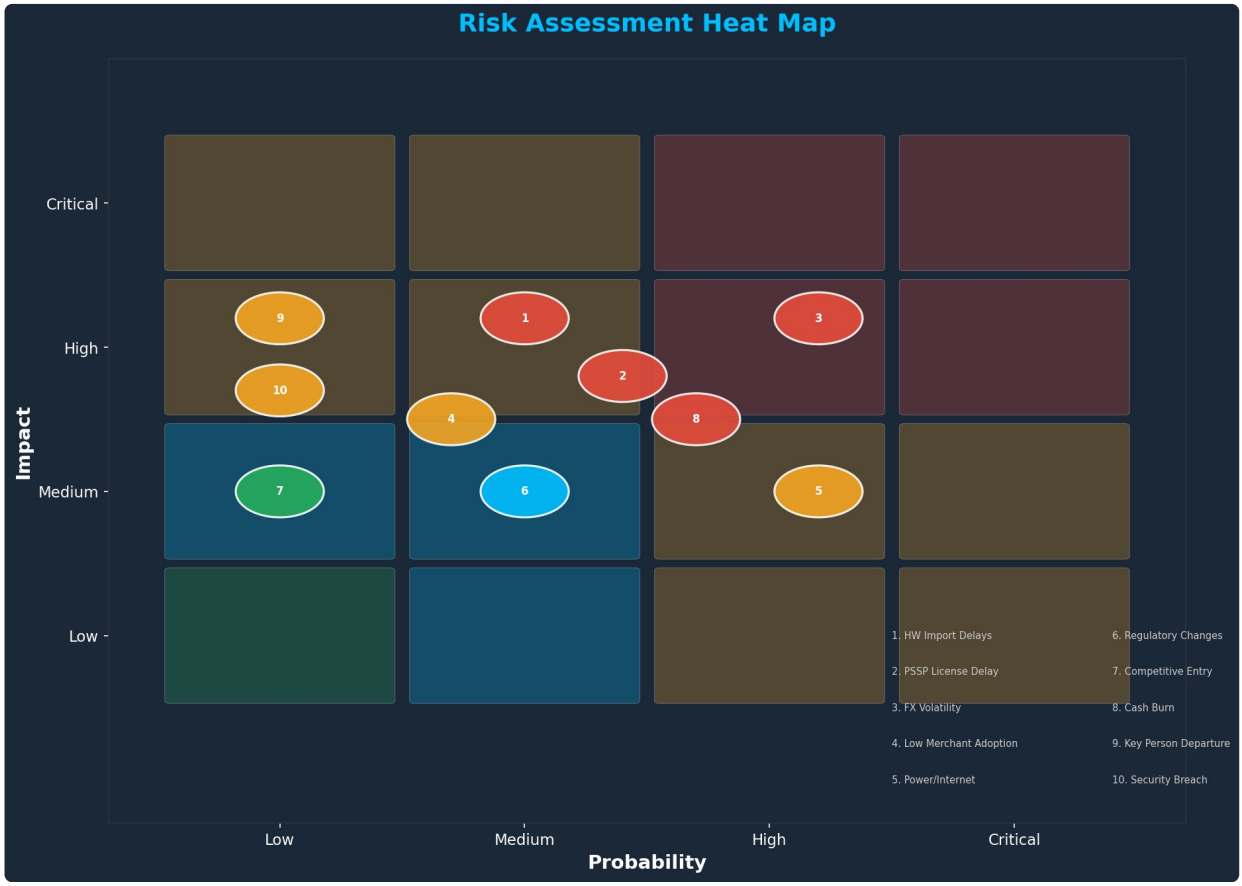

14. Risk Analysis & Mitigation



Risk Assessment Heat Map

#	Risk	Impact	Probability	Mitigation
1	**Hardware import delays**	**High**	**Medium**	**Order 3 months ahead; dual suppliers in Shenzhen**
2	**PSSP license delay**	High	Medium	Partner with Paystack/Flutterwave until approved
3	**FX volatility (Naira/USD)**	High	High	Pre-purchase hardware in bulk; hedge with USD reserves
4	**Low merchant adoption**	High	Medium	Free ERP + pilot incentives + market association partnerships
5	**Power/internet instability**	Medium	High	Offline-first design; power kits bundled; edge processing
6	**Regulatory changes (NDPA/CBN)**	Medium	Medium	Legal retainer; proactive compliance; data localization

7	**Competitive entry (Tuya/global)**	Medium	Low	Vertical specialization; African-optimized; installer loyalty
8	**Cash burn exceeds plan**	High	Medium	Conservative hiring; revenue milestones gate next phase
9	**Key person departure (CTO)**	High	Low	Equity vesting; knowledge documentation; pair programming
10	**Security breach**	High	Low	Zero-trust architecture; Keycloak auth; regular audits

Kill-Switch Rules

- **Do NOT** enter new city without 1 anchor client + local partner
- **Do NOT** launch new vertical without clear revenue owner
- **Do NOT** hire ahead of revenue — every new hire must be justified by growth metrics
- **STOP** hardware orders if sell-through rate drops below 60%

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15. Equity & Governance

15.1 Proposed Equity Structure

Stakeholder	Allocation	Notes
Founder(s)	**55-60%**	**Controlling stake, hardware + vision**
ESOP Pool	10–12%	CTO, Payments PM, key hires (4-year vesting)
Seed/Angel Investors	15–20%	This round (₦250M valuation basis)
Strategic Reserve	10–15%	Future rounds, partnerships

15.2 Governance

Position	Responsibility
Founder/CEO	**Vision, hardware sourcing, partnerships, capital**
CTO	Architecture authority, technical hiring, security
Advisory Board	Regulatory, finance, market access (3-5 advisors)

15.3 Investor Protection

- Monthly financial reporting
- Board seat for lead investor (optional)
- Quarterly milestone reviews
- Anti-dilution provisions on next round
- Right of first refusal on Series A

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16. Key Performance Indicators

16.1 Company-Level KPIs (Monthly Board Review)

KPI	Month 6 Target	Month 12 Target	Month 18 Target
Active Merchants (ERP)	**200**	**1,500**	**3,000**
IoT Devices Deployed	50	300	800
Markets Digitized	1	3	5
Spoke Stations Active	2	10	20
Monthly Revenue (₦)	1.5M	9.0M	14.0M
Payment GMV (₦/month)	20M	150M	500M
Team Size	12	25	35
Device Sell-Through Rate	70%	80%	85%
Customer Churn (monthly)	<15%	<10%	<8%
System Uptime	95%	99%	99.5%

16.2 Role-Specific KPIs

Role	Key Metrics
CTO	**System uptime, deployment frequency, zero security incidents**
Engineers	Features shipped/sprint, bug rate, code review quality
Market Ops	Merchants onboarded, active %, association satisfaction
Support	First-response time <2hrs, resolution <24hrs, churn reduction
Field Agents	15-25 shops/day, >98% data accuracy, <5% rejections

17. Use of Funds Summary

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┌──────────────────────────────────────────────────────────────────┐
│          ₦250,000,000 CAPITAL DEPLOYMENT                         │
│                  PECH Group Holdings Ltd                          │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  PEOPLE (28.0%)           ₦70,000,000                            │
│  ├─ Engineering Team         ₦32,000,000                         │
│  ├─ Operations & Support     ₦18,000,000                         │
│  ├─ Field Agents             ₦10,000,000                         │
│  └─ Contractors (UI/UX etc)  ₦10,000,000                         │
│                                                                  │
│  HARDWARE — 23 SKUs (30.0%) ₦75,000,000                          │
│  ├─ Commerce (POS, Tablets, Kiosks)  ₦24,000,000                 │
│  ├─ IoT & Energy Devices            ₦15,000,000                  │
│  ├─ Smart Water Controllers          ₦9,000,000                  │
│  ├─ Smart Home (Switches, Cameras)   ₦7,000,000                  │
│  ├─ Learning Tablets + LED Displays  ₦7,000,000                  │
│  ├─ Energy Storage Systems           ₦5,500,000                  │
│  └─ Shipping + Import + SONCAP      ₦7,500,000                   │
│                                                                  │
```

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│  CONTINGENCY & FX (8.4%)   ₦21,000,000                          │
│  └─ FX hedging, emergency, expanded product line buffer         │
│                                                                   │
│  TECHNOLOGY (9.6%)        ₦24,000,000                            │
│  ├─ Cloud Infrastructure       ₦17,100,000                       │
│  └─ AI Tools & Subscriptions   ₦7,200,000                        │
│                                                                   │
│  MARKETING (8.8%)         ₦22,000,000                            │
│  ├─ Digital Marketing          ₦8,000,000                        │
│  ├─ Field Marketing & Events   ₦7,000,000                        │
│  ├─ Brand & Content            ₦4,000,000                        │
│  └─ Demo Stations & Workshops  ₦3,000,000                        │
│                                                                   │
│  OPERATIONS (6.4%)        ₦16,000,000                            │
│  ├─ Office Rent (18 months)    ₦9,000,000                        │
│  ├─ Equipment & Furniture      ₦4,000,000                        │
│  └─ Utilities & Misc           ₦3,000,000                        │
│                                                                   │
│  LOGISTICS SETUP (4.8%)   ₦12,000,000                            │
│  ├─ Spoke Station Equipment    ₦8,000,000                        │
│  └─ Hub Setup & Branding       ₦4,000,000                        │
│                                                                   │
│  COMPLIANCE (4.0%)        ₦10,000,000                            │
│  ├─ SONCAP (3 shipments)       ₦4,500,000                        │
│  ├─ Legal Retainer (18mo)      ₦1,800,000                        │
│  ├─ Licenses & Permits         ₦2,550,000                        │
│  └─ Buffer                     ₦1,150,000                        │
│                                                                   │
│  [!] PSSP/PTSP (NEXT TRANCHE — NOT IN THIS BUDGET)               │
│  ├─ PSSP: Capital + Escrow     ₦205,000,000                      │
│  ├─ PTSP: Capital + Escrow     ₦201,000,000                      │
│  └─ PCI DSS + Trademarks       ₦7,000,000                       │
│                                                                   │
├──────────────────────────────────────────────────────────────────┤
│  TOTAL: ₦250,000,000                                             │
└──────────────────────────────────────────────────────────────────┘
```

18. Appendix

A. Open-Source Technology Stack (Licensing Summary)

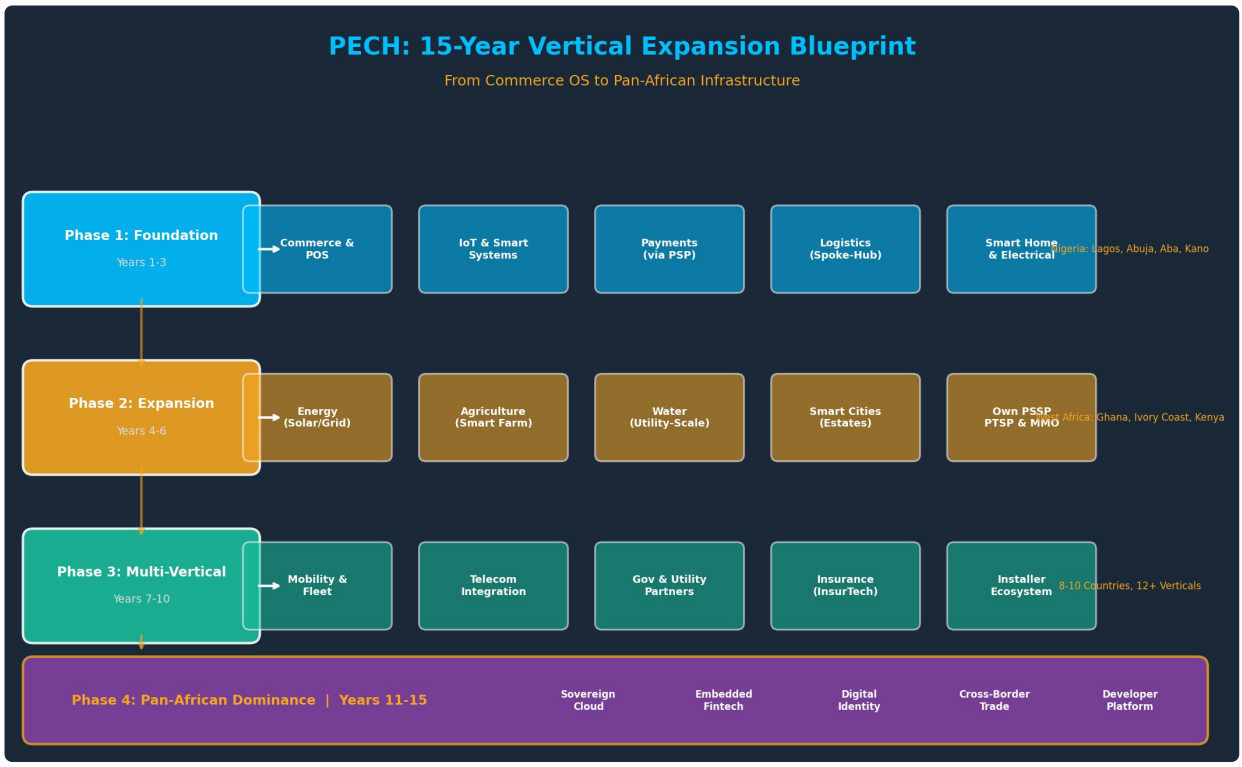
Technology	License	Commercial Use	Role in PECH
ThingsBoard	**Apache 2.0**	**Yes — rebrand OK**	**IoT platform base**
ERPNext	GPL v3	Yes (with compliance)	ERP base (fork)
EMQX	Apache 2.0	Yes	MQTT messaging
NanoMQ	MIT	Yes	Edge MQTT broker
Apache Kafka	Apache 2.0	Yes	Event streaming
Keycloak	Apache 2.0	Yes	Authentication
React	MIT	Yes	Web dashboards
Flutter	BSD	Yes	Mobile apps
MLflow	Apache 2.0	Yes	ML experiment tracking
Traefik	MIT	Yes	API gateway

k3s	Apache 2.0	Yes	Lightweight Kubernetes
Leaflet	BSD-2	Yes	Map visualization

B. Competitor Landscape

Competitor	Strengths	Weakness vs PECH
Tuya	**Global scale, cheap devices**	**No African localization, WiFi-dependent**
Paystack	Payment dominance	No ERP, no hardware, no IoT
TradeDepot	Distribution network	No IoT, no marketplace, no platform
Moniepoint	POS dominance	No ecosystem, no IoT, no marketplace
ThingsBoard	Open-source IoT	No commerce, no payments, no hardware

C. 15-Year Vertical Expansion Blueprint



15-Year Expansion Blueprint

Phase	Years	Geography	Verticals Added	Key Milestones
Foundation	**1–3**	**Nigeria (Lagos, Abuja, Aba, Kano)**	**Commerce & POS, IoT & Smart Systems, Payments (via PSP), Spoke-Hub Logistics, Smart Home & Electrical, Smart Water Controllers, Learning Tablets, Energy Storage**	**ERP at 3,000+ merchants, 800+ IoT devices, own PSSP/PTSP licenses obtained**
Expansion	4–6	West Africa (Ghana,	Energy (Solar/Mini-Grid), Agriculture (Smart Farming),	Multi-country operations, 10+

		Ivory Coast, Kenya)	Water (Utility-Scale), Smart Cities (Estate Mgmt), Own PSSP/PTSP/MMO	verticals, installer ecosystem launched
Multi-Vertical	7–10	8–10 African countries	Mobility & Fleet Tracking, Telecom Integration, Government & Utility Partnerships, Insurance (InsurTech), Certified Installer Network	Cross-border trade platform, government contracts, 12+ verticals
Pan-African Dominance	11–15	15+ countries	Sovereign Cloud Infrastructure, Embedded Fintech (Loans/Credit), Digital Identity, Cross-Border Trade Platform, Developer Platform & Open API Ecosystem	Pan-African infrastructure OS, developer marketplace, data services

Note: Revenue projections for years 4-15 are not included as they depend on market conditions, regulatory environments, and capital availability at each phase. PECH will publish updated projections at the end of each phase based on actual performance data.

D. Contact

PECH Group Holdings Ltd Lagos, Nigeria [Contact details to be inserted]

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